

The $\mathrm{RCD}(K, \infty)$ Log-Sobolev Rigidity Problem Was Answered by Han

Math discovery packet

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Status and source problem

This is a lightweight *literature-already-answered* status note, not an original proof. Ohta and Takatsu, arXiv:1904.09400, Section 4 (“Further problems”), item (B), page 10 of the locally compiled source PDF, write:

Cheng–Zhou’s characterization of the sharp spectral gap was extended to $\mathrm{RCD}(K, \infty)$ -spaces. Hence it is natural to expect the analogue to Theorem 1.1 for $\mathrm{RCD}(K, \infty)$ -spaces. They suggest either extending needle decomposition to this setting or directly showing that $\log \rho$ attains the sharp spectral gap.

Thus the requested analogue says that a nonconstant equality case in the sharp logarithmic Sobolev inequality, under $K > 0$, forces a one-dimensional Gaussian factor and that the extremal density is a translation along that factor.

The later answer

Bang-Xian Han, *Rigidity of some functional inequalities on RCD spaces*, arXiv:2001.07930, Theorem 1.7 (page 8 of the locally compiled supporting PDF), proves the following. If an $\mathrm{RCD}(K, \infty)$ probability space with $K > 0$ admits a nonconstant equality case for an admissible Φ -entropy inequality, then it splits isometrically as

$$\left(\mathbb{R}, |\cdot|, \sqrt{\frac{K}{2\pi}} e^{-Kr^2/2} dr \right) \times (Y, d_Y, m_Y).$$

For $\Phi(x) = x \log x$, the logarithmic-Sobolev extremal is

$$f(r, y) = \exp\left(ar - \frac{a^2}{2K}\right), \quad a \in \mathbb{R},$$

which is precisely the Radon–Nikodym density of a translation along the Gaussian coordinate. This gives both conclusions of the desired analogue: Gaussian splitting and classification of equality functions.

Why the provenance is explicit

Han’s Remark 1.9 (page 9) cites Ohta–Takatsu and states that their Section 4 left rigidity of the log-Sobolev inequality on $\text{RCD}(K, \infty)$ spaces open because dimension-free needle decomposition was unavailable. The remark then says explicitly that Han’s result extends Ohta–Takatsu to this setting. The supporting author therefore knew that the theorem answered the same problem.

Scope

The status is `literature_already_answered` (Further problem (B), affirmative). Items (A), (C), (D), and (E) in the source’s Section 4 are not assessed here.

References

- S.-i. Ohta and A. Takatsu, *Equality in the logarithmic Sobolev inequality*, arXiv:1904.09400, Section 4(B).
- B.-X. Han, *Rigidity of some functional inequalities on RCD spaces*, arXiv:2001.07930, Theorem 1.7 and Remark 1.9.